

BULLET-LA FRETAZ, SwitzerlandWMO: **066190**Lat: **46.8406N**Lon: **6.5764E**Elev: **1210**StdP: **87.61**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.6	-9.8	-17.2	1.0	-4.6	-14.3	1.3	-4.6	11.2	1.8	9.3	0.5	2.8	60	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	24.1	16.8	22.5	16.2	21.0	15.5	17.8	22.4	16.9	21.2	16.1	20.2	2.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	13.3	20.3	15.2	12.5	19.3	14.4	11.8	18.4	55.4	22.5	52.4	21.3	49.6	20.2	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.3	5.3	DB	-13.1	26.7	2.8	1.6	-15.1	27.9	-16.7	28.8	-18.3	29.7	-20.3	30.9	(4)
(5)			WB	-13.4	19.8	2.7	1.1	-15.3	20.6	-16.9	21.2	-18.4	21.8	-20.4	22.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-1.2	-1.3	2.0	5.9	9.0	13.4	15.2	14.7	11.2	8.0	3.0	0.1	(6)	
	DBStd	7.19	4.25	4.83	4.46	4.24	4.00	4.07	3.58	3.69	3.45	4.20	4.50	4.41	(7)	
	HDD10.0	1769	348	318	249	134	68	13	3	4	27	87	211	307	(8)	
	HDD18.3	4276	606	551	507	373	289	154	108	125	216	322	459	565	(9)	
	CDD10.0	564	0	0	1	11	37	115	165	148	62	24	1	0	(10)	
	CDD18.3	30	0	0	0	0	1	7	12	11	0	0	0	0	(11)	
	CDH23.3	82	0	0	0	0	1	19	32	31	0	0	0	0	(12)	
	CDH26.7	5	0	0	0	0	0	1	2	2	0	0	0	0	(13)	
(14) Wind	WSAvg	2.1	2.2	2.3	2.4	2.3	2.2	2.1	2.0	1.8	1.9	1.9	2.1	2.3	(14)	
(15) Precipitation	PrecAvg	1284	102	95	88	97	111	110	114	110	106	114	117	121	(15)	
	PrecMax	1633	274	256	338	170	188	238	246	220	240	210	323	254	(16)	
	PrecMin	917	19	19	24	16	37	47	52	27	40	24	12	0	(17)	
	PrecStd	190	60	57	68	49	46	47	47	49	50	51	70	63	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	10.7	13.0	17.8	21.2	25.5	26.4	26.3	21.0	18.1	13.7	10.5	(19)	
		MCWB	3.0	4.2	7.1	10.6	14.7	18.0	17.4	16.9	15.3	12.1	8.3	3.7	(20)	
	2%	DB	7.7	8.2	11.1	15.6	18.8	23.3	24.0	23.8	19.2	16.2	12.0	8.5	(21)	
		MCWB	3.3	2.8	5.8	9.3	13.3	16.9	16.5	16.7	14.7	11.1	7.3	3.4	(22)	
	5%	DB	5.9	6.3	9.6	13.8	17.0	21.4	22.3	22.1	17.6	14.8	10.5	7.1	(23)	
		MCWB	3.1	2.3	5.2	8.5	11.8	15.8	16.1	16.1	13.9	10.6	6.5	3.4	(24)	
	10%	DB	4.5	4.7	8.1	12.1	15.2	19.7	20.7	20.3	16.2	13.3	9.1	5.7	(25)	
		MCWB	2.4	2.0	4.3	7.7	10.9	14.9	15.3	15.4	13.0	9.9	6.0	2.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	5.5	8.3	11.3	15.6	19.5	18.7	18.8	16.8	13.6	10.0	6.7	(27)	
		MCDB	7.1	7.9	10.7	15.9	19.8	24.0	23.5	23.4	19.4	16.4	12.2	7.6	(28)	
	2%	WB	4.9	4.2	7.0	10.1	13.9	17.7	17.5	17.6	15.4	12.3	8.4	5.4	(29)	
		MCDB	6.0	6.5	10.0	14.6	17.9	22.1	22.2	22.1	18.1	14.5	10.4	6.7	(30)	
	5%	WB	3.8	3.3	5.9	8.9	12.5	16.4	16.7	16.7	14.3	11.5	7.4	4.5	(31)	
		MCDB	5.1	5.4	8.7	13.0	16.0	20.5	21.2	20.9	16.9	13.6	9.5	6.0	(32)	
	10%	WB	2.6	2.3	4.9	7.9	11.3	15.2	15.8	15.8	13.4	10.7	6.5	3.6	(33)	
		MCDB	4.1	4.4	7.4	11.7	14.6	19.0	20.1	19.7	15.8	12.7	8.5	5.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	4.8	5.5	6.4	6.6	7.0	7.2	6.6	5.9	5.4	4.8	4.6	(35)	
		MCDBR	5.9	6.4	7.3	8.2	8.6	8.5	8.6	8.1	7.4	7.1	6.3	5.9	(36)	
	5% WB	MCWBR	4.0	4.5	5.0	5.2	5.8	5.8	5.1	5.3	5.2	5.1	4.4	4.0	(37)	
		MCWBR	5.2	5.9	6.4	7.7	8.0	8.1	8.0	7.7	6.7	6.1	5.8	5.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.262	0.282	0.312	0.342	0.350	0.352	0.351	0.345	0.328	0.312	0.284	0.261	(40)	
		taud	2.488	2.450	2.439	2.381	2.393	2.415	2.405	2.437	2.472	2.512	2.535	2.502	(41)	
	Ebn at Noon		854	900	917	913	914	911	906	900	888	852	818	814	(42)	
		Edh at Noon	63	81	96	112	115	113	112	105	92	76	61	56	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.28	2.13	3.29	4.36	5.00	5.93	5.78	5.03	3.91	2.50	1.38	1.04	(44)	
	RadStd		0.15	0.32	0.48	0.63	0.45	0.56	0.50	0.46	0.35	0.25	0.18	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (94 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
						+0.39	+0.29				

Nomenclature: See separate page